

SugarCluster — Progress Summary

Course: CPSC 4520 — Distributed Systems Date: May 30, 2026

Overview

For this project we're building middleware to run the Sugarscape agent-based simulation at scale on the ACES supercomputer. The goal is to study how disease parameters and ethical frameworks affect disease spread. We're sweeping across 8 ethical frameworks and 81 disease parameter combos, plus 8 baselines — 656 simulation jobs total.

What We Have Working

Config generator (`generate_configs.py`): Reads `sweep.toml`, generates 656 JSON configs and a `jobs.csv` manifest. Each config maps to a SLURM array task. Supports 4 disease parameters (transmission chance, tag string length, immune system length, metabolism penalty) with 3 values each.

SLURM submission (`submit.slurm`): Array job that runs `--array=1-656%50`, 1 hour walltime, 4GB memory per task. Reads the config path from `jobs.csv` and runs Sugarscape headless.

ACES setup script (`setup_aces.sh`): Handles Python 3.12 detection, venv creation, `tomli` install, and runs a smoke test to make sure everything works.

Output checker (`check_outputs.py`): Goes through all 656 logs, checks they exist, are valid JSON, aren't empty, and ran to the full 1000 timesteps. Writes failed jobs to `retry.csv`.

Local dry-run: Tested a 5-timestep run locally. Sugarscape loads the config fine and produces valid JSON output.

What's Not Done Yet

- `main.py` is just a placeholder — no orchestration logic
- No data analysis code — need `aggregate.py` to parse logs and `analyze.py` for statistics
- No plotting or visualization scripts
- No README, slides, or final submission package

Remaining Work

Phase	What	Status
1	Transfer to ACES, run setup, pilot job	Not started
2	Submit all 656 jobs, monitor, validate	Not started
3	Pull logs back, write aggregate/analyze scripts	Not started
4	Generate plots, answer research questions	Not started
5	Slides, README, Makefile, <code>final.zip</code>	Not started

Next up is getting the code onto ACES and running a pilot job to make sure the whole pipeline works end-to-end. Plan to submit all 656 jobs tonight and let them run overnight.